



UNITED INTERNATIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT PROPOSAL

HaluRISC: Calibrated and Explainable Hallucination Risk Prediction for Large Language Model Outputs

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1 Project Overview

Large language models (LLMs) like ChatGPT are very good at writing text that sounds confident and correct, but they sometimes generate information that is wrong or made up — this is called a hallucination. This project builds a complete system that predicts how likely an LLM answer is to be hallucinated. The system is black-box friendly — it only needs the question, the answer, and (optionally) some context, without ever looking inside the LLM. It is lightweight — it uses simple text features and classical machine learning (XGBoost), so a full analysis runs in about 62ms at the median on a laptop, with no API fees. It is explainable — for every prediction it shows *why* the answer looks risky (for example: “the answer contradicts the provided context”, or “the answer contains numbers that are not supported”). Finally, it is presentable — a polished, interactive web dashboard lets a judge or supervisor test the system live, type any question and answer, and see the risk score and explanation instantly, with every claim in the answer verified against the pasted context, uploaded documents, or live web search, and cited. The novelty is not in inventing hallucination detection, but in combining NLI-based consistency features, calibrated risk scores, and tested explanations into one lightweight system — a combination no published work offers as a single product.

2 Motivation

LLMs are now used in real products — chatbots, Q&A systems, content review tools — where a wrong answer has a real cost, and the most common way to detect hallucinations is to ask the LLM itself to judge its own output [2], which is slow, expensive, and sometimes wrong. A lightweight model trained on simple text features (word overlap, named entities, contradiction with context, numbers, hedging words) can flag risky answers almost instantly and at almost zero cost. A risk score is also more useful than a simple “yes/no” flag, because real systems can decide what to do with a risky answer — show a warning, ask for human review, or fetch better evidence. And because the final result is a working product with a clean UI rather than just a report, the project is fully ready for live demonstration.

3 Objectives

Research question: Can a lightweight machine learning model predict hallucination risk in LLM answers using simple text features, while giving calibrated risk scores and clear explanations? To answer this, the project has five objectives. The first is to build a feature extraction pipeline (about 20–30 features) that captures how well an answer agrees with the question and context, covering word overlap, named entities, contradiction signals, numbers, hedging words, and semantic similarity. The second is to train and honestly compare classical models, including a simple heuristic baseline, Logistic Regression, Random Forest, and XGBoost. The third is to calibrate the risk scores so that a score of 0.8 truly means “about 80% risk”, making the score trustworthy for real use. The fourth is to explain predictions with SHAP, giving a global view of which features matter most plus detailed case studies of individual predictions. The fifth is to build a polished web dashboard (Next.js 16 with assistant-ui + FastAPI) with strong UI/UX, featuring live prediction, a risk gauge, per-claim evidence verdicts, an explanation panel, and a demo gallery, designed for presentation and judging.

4 Related Work

Hallucination detection is an active research area. The most common approach asks the LLM itself to judge its own answer, as in SelfCheckGPT [2], but this is slow and expensive. To cut

this cost, small language models have been used as detectors [9], and lightweight evaluation models such as Luna [10] show that efficient detection is possible in practice. Other approaches train neural classifiers on frozen language encoders [7], or combine explainable text signals with machine learning [6, 8].

On the benchmark side, HaluEval [1] and TruthfulQA [3] provide standard test beds, and fine-grained methods such as FActScore [4] evaluate factual precision in detail. RAGTruth [13] adds natural responses with human word-level annotations. Newer benchmarks such as FaithBench [11] show that current detectors still fail on challenging cases, and SpikeScore [12] shows that cross-domain generalization remains an open problem.

None of these works combine lightweight engineered features with calibrated risk scores, tested explanations, and a deployable dashboard — this is the gap the project fills.

5 Methodology

The pipeline has seven clear steps:

1. Data preparation — load the HaluEval QA dataset, clean the text, and manually inspect 50 random samples to verify label quality before training.
2. Feature extraction — compute features in seven groups: length/style, lexical overlap, entity overlap, NLI contradiction (using a lightweight pre-trained NLI model), numeric consistency, hedging phrases, and semantic similarity.
3. Model training — 5-fold stratified cross-validation, hyperparameter tuning for XGBoost, and every experiment repeated with 3 random seeds, reported as mean $\pm$ standard deviation.
4. Model comparison — heuristic baseline, Logistic Regression, Random Forest, and XGBoost, judged on F1, AUROC, precision, recall, and MCC.
5. Calibration — compare Platt scaling and isotonic regression fit on the validation split only [5], measured with Expected Calibration Error (ECE) and Brier score, and re-calibrate on target-domain data when a small labeled sample is available.
6. Explanation — SHAP global feature importance plus 3 local case studies; error analysis on 20 wrong predictions (10 false positives, 10 false negatives).
7. Deployment — FastAPI backend serving the trained model, and a Next.js 16 + assistant-ui dashboard with a clean, modern UI: risk gauge, feature breakdown, per-claim evidence verdicts with citations, and a live input panel.

Statistical rigor: model differences are tested with McNemar’s test and bootstrap confidence intervals, so conclusions are not based on chance from a single split.

External comparison: results are compared with published HaluEval detection baselines, such as SelfCheckGPT [2], and validated on the RAGTruth corpus [13], so a judge can see whether the scores are good and whether they generalize.

6 Dataset

We use the HaluEval benchmark [1] (EMNLP 2023) — the standard dataset for hallucination evaluation, widely used and well documented.

- Size: 35,000 samples across 4 task types (QA, dialogue, summarization, general).
- Used here: the QA subset (about 10,000 samples). Each sample has a question, an answer, a context, and a label (hallucinated or not).
- Why QA: it is the largest, cleanest, and most comparable subset, and it matches the demo

scenario (question $\rightarrow$ answer $\rightarrow$ risk score).

- External validation: results will also be checked on RAGTruth [13] — 17,790 naturally generated responses from several modern LLMs with human word-level annotations — to test whether the model generalizes to natural responses.

7 Results

On the HaluEval QA subset (20,000 rows, 70/15/15 grouped stratified split) the tuned XGBoost model reaches $F1 = 0.9846$, $AUROC = 0.9979$, and $MCC = 0.9697$, with an expected calibration error of 0.0045 before any recalibration. The full analysis runs in about 62 ms at the median and costs about \$0.001 per 1,000 predictions, roughly 100 times cheaper than an LLM-as-judge that it outperforms on a measured 200-sample set ($F1$ 0.985 vs. 0.841). Zero-shot transfer to natural RAGTruth responses is poor, with $F1 = 0.3022$ on the QA split, an honest negative result that is reported rather than hidden. A small target-domain recalibration on 5,034 RAGTruth rows cuts the ECE on that corpus from 0.8185 to 0.1335. The project delivers a calibrated, explainable, deployable system with claim-level verification, evidence citations, and a live web dashboard.

The project will deliver:

1. A trained, calibrated XGBoost model with feature ablation, SHAP explanations, and explanation-reliability measurements.
2. Statistical significance tests (McNemar, bootstrap confidence intervals, three-seed protocol).
3. A polished, judge-ready web dashboard — live risk prediction, per-claim evidence verdicts with citations, and an experiment dashboard fed by the real artifacts.
4. Reproducible code and artifacts: a self-contained Colab notebook, a frozen manifest with hashes, and the complete pipeline (data, features, training, evaluation, API).

8 Project Plan

The project ran in two phases — experiments first, then the product — and followed the pipeline below.

- Data preparation: downloaded the HaluEval QA subset, cleaned the text, and manually audited 50 random samples to verify label quality.
- Feature extraction: computed 26 features in seven groups — length/style, lexical overlap, entity overlap, NLI contradiction, numeric consistency, hedging phrases, and semantic similarity.
- Model development: compared a heuristic baseline, Logistic Regression, Random Forest, and XGBoost using grouped 5-fold cross-validation, randomized hyperparameter search, and 3-seed runs.
- Calibration and evaluation: compared Platt and isotonic calibration, ran McNemar and bootstrap significance tests, evaluated zero-shot transfer on RAGTruth [13] and FaithBench [11], and measured explanation reliability with perturbation analysis.
- Explanation: produced SHAP global feature importance, 3 local case studies, and an error analysis of 48 misclassified test samples.
- Product: built the FastAPI backend and the Next.js + assistant-ui dashboard with claim-level verification, document indexing, live web search, and a demo page.
- Documentation: prepared the final report and rehearsed the live demo.

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